

SDLC MODEL SELECTION REPORT

1. Introduction

The selected project for this analysis is an **Online Shopping Website (E-commerce Platform)**. An online shopping website is a complex, dynamic web application that allows users to browse products, add items to a cart, and securely complete transactions using various payment gateways. The platform must incorporate robust user authentication, inventory management, a seamless user interface (UI), and high-security standards to protect sensitive financial data. Because the e-commerce landscape is highly competitive, the website must continually adapt to user trends, promotional seasons, and technological advancements.

2. Selected SDLC Model

Selected Model: Agile SDLC (Scrum Framework)

The most appropriate Software Development Life Cycle (SDLC) model for an online shopping website is the **Agile Model**. Agile focuses on iterative development, continuous feedback, and rapid delivery of functional software components. By breaking the project down into short development cycles called "Sprints", the development team can frequently release functional modules (e.g., product catalog first, payment integration next).

3. Justification

The Agile model is highly suited for this project due to the following critical reasons:

- **Requirement Stability (Highly Dynamic):** E-commerce requirements frequently change based on market trends, competitor actions, or new compliance regulations. Agile embraces changing requirements, even late in development, ensuring the product remains highly relevant upon launch.
- **Risk Level (High Business Risk):** Launching an unverified e-commerce platform poses a high financial risk. Agile mitigates this by delivering a Minimum Viable

Product (MVP) early. Early launches allow stakeholders to test core functions (like browsing and checkout) and fix critical issues before full-scale deployment.

- **Complexity (High Integration Needs):** Developing an online store involves complex integrations: CRM systems, payment APIs, shipping calculators, and inventory databases. Agile's iterative approach allows developers to tackle, test, and validate these complex integrations one at a time rather than dealing with all of them in a massive final integration phase.
- **User Involvement (Continuous Feedback):** The success of an online shopping website relies heavily on the User Experience (UX). Agile requires continuous stakeholder and beta-tester feedback after every sprint, allowing the team to fine-tune the user interface, navigation flow, and checkout process based on real user interactions.
- **Flexibility & Time-to-Market:** Promotions, holidays, or sudden market demands require quick feature additions (e.g., adding a "Black Friday Sale" module). Agile provides the flexibility to reprioritize the product backlog and deliver high-priority features rapidly without derailing the entire project timeline.

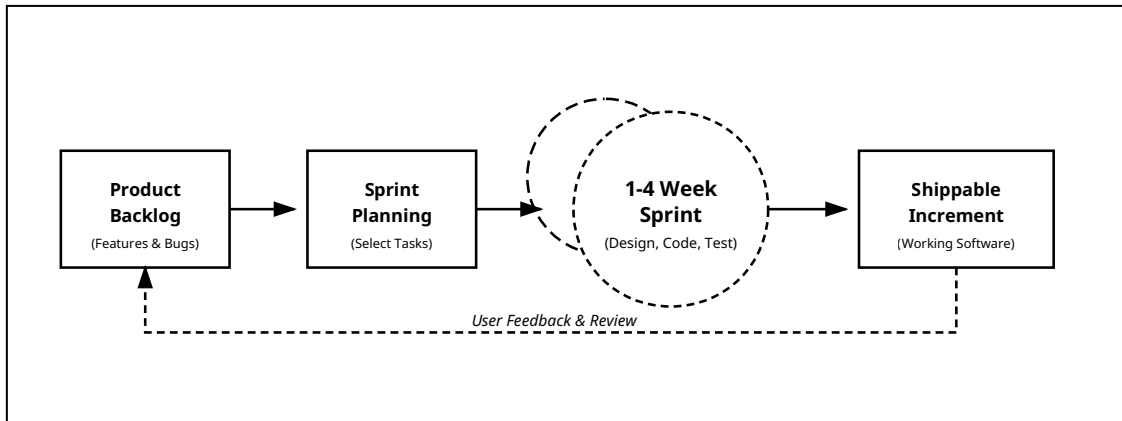
4. Comparison with Other Models

Alternative SDLC models are significantly less effective for an online shopping website for the following reasons:

- **Waterfall Model:** The Waterfall model requires all requirements to be gathered at the beginning and strongly discourages changes once development starts. This rigidity is fatal for an e-commerce platform where market trends evolve rapidly. Furthermore, testing happens only at the very end, meaning critical UX flaws or payment integration bugs might not be discovered until the product is nearly finished.
- **V-Model:** While the V-Model offers strict validation and verification, it is heavily structured and lacks flexibility. It is better suited for projects with static requirements and mission-critical systems (like aerospace or healthcare equipment). For an e-commerce site, the inability to pivot features easily makes the V-Model overly restrictive and costly.

5. Agile SDLC Diagram

The diagram below illustrates the iterative loop of the Agile Scrum methodology, showing how an E-commerce platform is built incrementally.



6. Conclusion

In conclusion, developing an online shopping website requires a development methodology that embraces change, promotes rapid feature delivery, and prioritizes user satisfaction. The **Agile SDLC model** perfectly aligns with these necessities. By favoring iterative sprints and continuous feedback over rigid, predetermined phases, Agile minimizes the risk of product failure and ensures that the final e-commerce platform is robust, secure, and fully aligned with current market expectations.